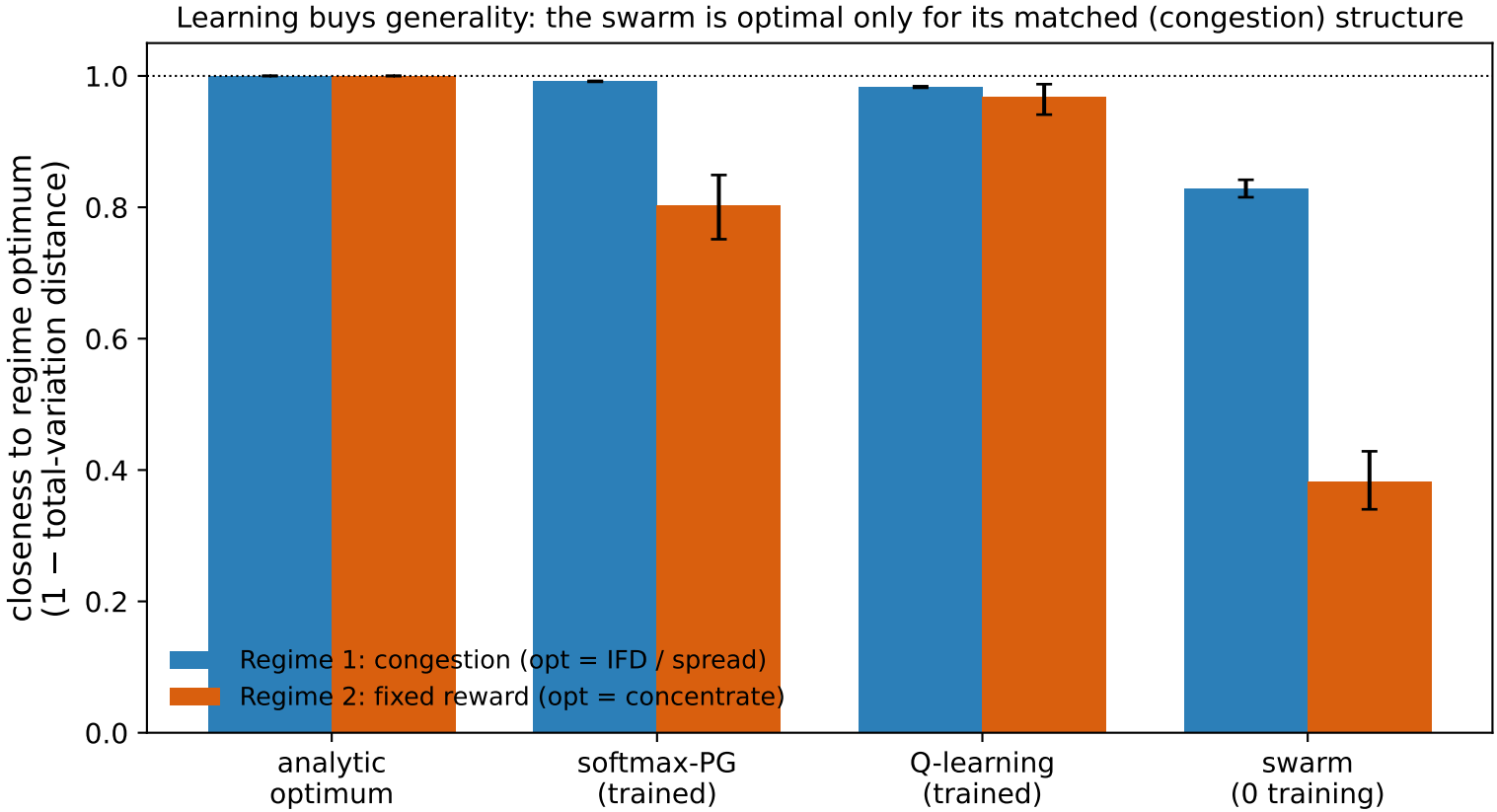
Learning buys generality: the swarm is optimal only for its matched (congestion) structure
closeness to regime optimum
(1 − total-variation distance)
1.0
0.8
0.6
0.4
0.2
0.0
Regime 1: congestion (opt = IFD / spread)
Regime 2: fixed reward (opt = concentrate)
analytic optimum
softmax-PG (trained)
Q-learning (trained)
swarm (0 training)